

Module 2: Transport Layer and Network Layer



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Hands-on

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4.4 Multicast Routing – Overview

- **Unicast:** one source to one destination.
- **Multicast:** one source to a group of destinations.
- **Broadcast:** one source to all hosts.
- This section covers multicast concepts, addressing, membership, forwarding, routing trees, intradomain and interdomain protocols.

Internet Routing

Multicast routing is needed when the same information must reach multiple receivers without creating independent copies at the source.

Unicasting, Multicasting and Broadcasting

Unicasting

One source → one destination. A router forwards through one interface.

Multicasting

One source → a group. A router may forward copies through several interfaces.

Broadcasting

One source → all hosts. Internet-wide broadcasting is not provided.

Figure 4.87

Figure 4.87 *Unicasting*

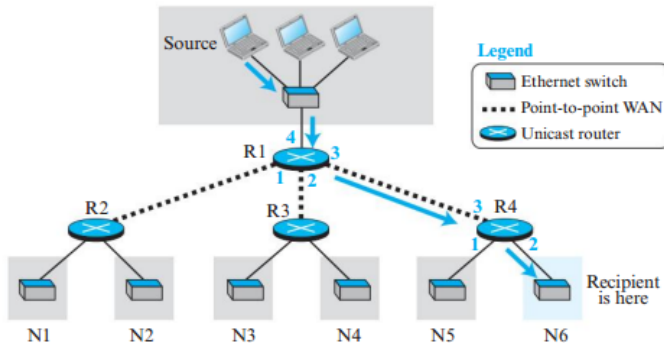
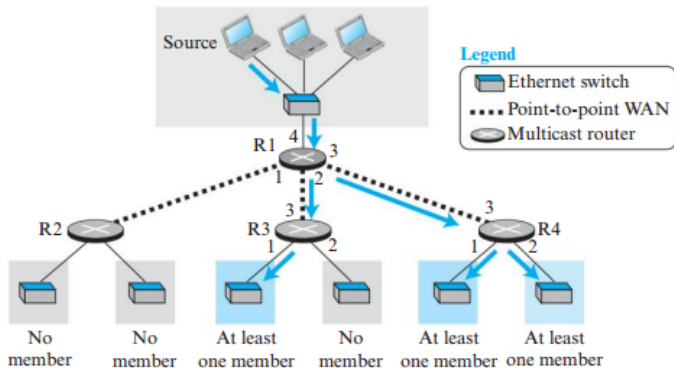


Figure 4.88

Figure 4.88 *Multicasting*



Multicasting vs. Multiple Unicasting

Multicasting

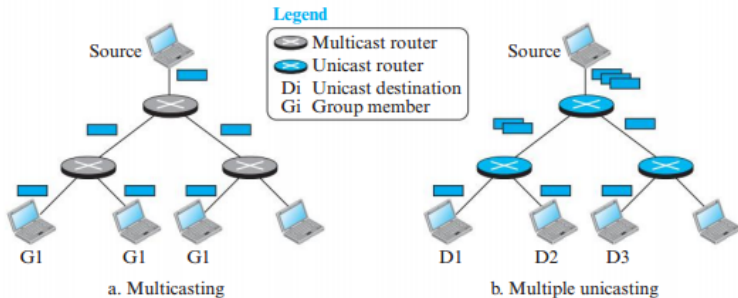
One packet starts at the source; routers duplicate it when required. All copies use the same multicast destination address. Only one copy travels between any two routers.

Multiple Unicasting

Several packets start at the source, each with a different unicast destination. Multiple copies may traverse the same link. Example: one e-mail sent separately to several recipients.

Figure 4.89

Figure 4.89 *Multicasting versus multiple unicasting*



Why Multicasting?

Bandwidth

Multicasting is more efficient than multiple unicasting because routers duplicate packets only where paths diverge.

Delay

With many destinations, multiple unicasting requires the source to create packets one after another. Multicasting creates only one source packet.

Multicast Applications

Distributed Databases

- Request can be multicast to database locations.

Information Dissemination

- Software updates, news and similar information.

Teleconferencing

- Same information reaches participants simultaneously.

Distance Learning

- One lesson can reach a selected student group.

Definition

Broadcasting is one-to-all communication: a host sends a packet to all hosts in an Internet.

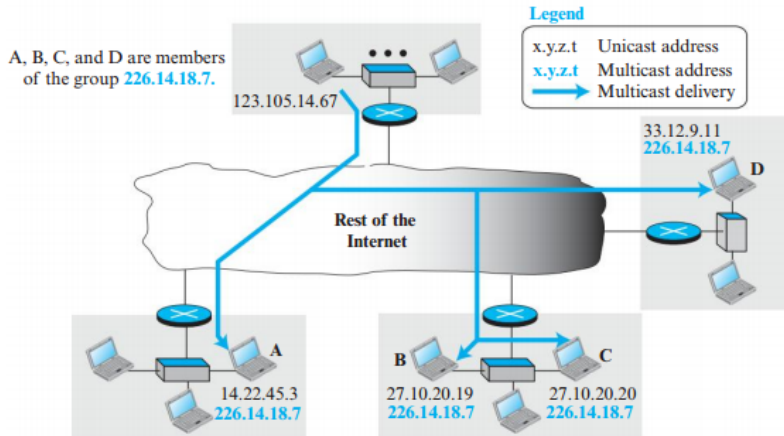
- Internet-wide broadcasting is not provided because it can create huge traffic and bandwidth usage.
- Partial or controlled broadcasting can occur within a domain.
- Some peer-to-peer applications may use broadcasting.

Multicast Addresses

- A multicast address identifies a **group of recipients**.
- Source address can remain a unicast address.
- Destination address is the multicast group address.
- A host belonging to n groups has $n + 1$ addresses: one unicast address and n multicast addresses.

Figure 4.90

Figure 4.90 Needs for multicast addresses



IPv4 Multicast Addresses

Multicast Block

224.0.0.0/4

Range: 224.0.0.0 to 239.255.255.255.

- Formerly Class D.
- Four bits identify the multicast block.
- Remaining bits identify the group.
- Number of addresses: 2^{28} .

Figure 4.91

Figure 4.91 *A multicast address in binary*



IPv4 Multicast Address Subblocks

Subblock	Range	Purpose
Local Network Control	224.0.0.0/24	Used inside a network; routers do not forward it
Internetwork Control	224.0.1.0/24	Multicast protocol use across the Internet
SSM	232.0.0.0/8	Source-Specific Multicast
GLOP	233.0.0.0/8	Range associated with an AS number
Administratively Scoped	239.0.0.0/8	Restricted to an administrative area

Selecting a Multicast Address

Limited Group

A group confined to an AS can use an unused address in the administratively scoped range $239.x.y.0-239.x.y.255$.

Example

AS number 23452 can be represented as $(91.156)_{256}$. A corresponding GLOP range is $233.91.156.0-233.91.156.255$.

Larger Group

For a group beyond an AS, the source-specific multicast block $232.0.0.0/8$ can be used.

Collecting Information About Groups

Unicast

- 1 Router knows attached networks automatically.
- 2 Routing protocols propagate this information.

Multicast

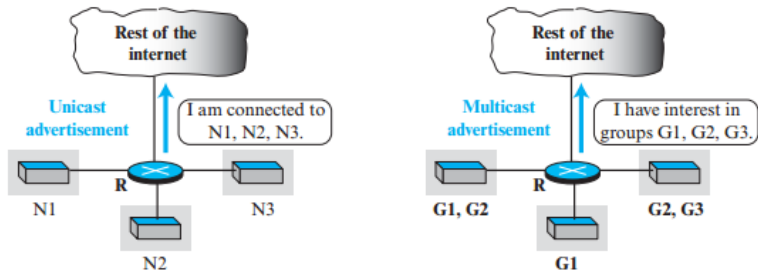
- 1 Router must discover active group members.
- 2 Membership can change dynamically.
- 3 Membership information is propagated by multicast routing protocols.

Key Difference

Multicast needs a mechanism to discover which groups are active on each interface.

Figure 4.92

Figure 4.92 *Unicast versus multicast advertisement*



Internet Group Management Protocol (IGMP)

- IGMP collects multicast group membership information.
- It is a network-layer auxiliary protocol associated with IP.
- IGMP messages are encapsulated in IP datagrams.

Query

General; group-specific;
source-and-group-specific.

Report

Sent by a host in response to a query and carries group/source membership information.

IGMPv3 Query and Report

Query types

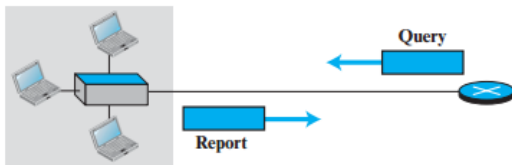
- General query: destination 224.0.0.1.
- Group-specific query: asks about one multicast group.
- Source-and-group-specific query: asks about a group with respect to specific source(s).

Report

- Contains group records and source inclusion/exclusion information.
- Encapsulated using multicast destination 224.0.0.22.

Figure 4.93

Figure 4.93 *IGMP operation*



IGMP Membership Changes

- To **join**, a host waits for a query and sends a report.
- To **leave**, a host does not respond to the corresponding query.
- If no other host responds, the router removes the group from its database.

Purpose

IGMP provides the membership information required by multicast routing.

Multicast Forwarding

Unicast

- Destination identifies one destination.
- Normally one outgoing interface.
- Decision depends on destination address.

Multicast

- Destination identifies a group.
- Multiple outgoing interfaces may be required.
- Decision depends on source and destination addresses.

Why Source Matters

The router must avoid sending a duplicate packet back toward an area that already received it.

Figure 4.94

Figure 4.94 *Destination in unicasting and multicasting*

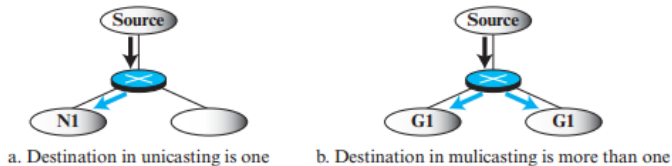
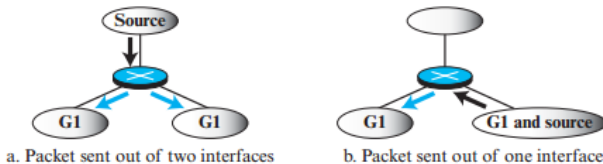


Figure 4.95

Figure 4.95 *Forwarding depends on the destination and the source*



Two Approaches to Multicasting

Source-Based Tree

Separate tree for every source-group combination. Source is root and group members are leaves.

$m \times n$ trees

for m groups and n sources.

Group-Shared Tree

A core/rendezvous point represents each group and is the root.

m trees

for m groups.

Group-Shared Tree and Tunneling

- 1 Source sends the multicast packet to the core using unicast tunneling.
- 2 Multicast packet is encapsulated in a unicast packet.
- 3 Core decapsulates it.
- 4 Core multicasts the original packet to group members.

Source → Core → Group Members

Trade-off

Fewer routing trees, but a mechanism is required to select the core.

Intradomain Multicast Routing Protocols

- ① **DVMRP** – extension of RIP; source-based tree.
- ② **MOSPF** – extension of OSPF; source-based tree.
- ③ **PIM** – uses an underlying unicast routing protocol.

PIM Modes

PIM-DM uses the source-based approach; PIM-SM uses the group-shared approach.

DVMRP – Distance Vector Multicast Routing Protocol

DVMRP extends RIP for multicast routing and uses source-based trees.

Three Steps

- ① Reverse Path Forwarding (RPF)
- ② Reverse Path Broadcasting (RPB)
- ③ Reverse Path Multicasting (RPM)

Reverse Path Forwarding (RPF)

- Router accepts a multicast packet only if it arrives through the interface on the shortest path back to the source.
- The router consults its unicast forwarding table as if it were sending a packet toward the source.
- Correct incoming interface $\Rightarrow$ accept.
- Wrong incoming interface $\Rightarrow$ discard.

RPF prevents loops and duplicate multicast packets

Reverse Path Broadcasting (RPB)

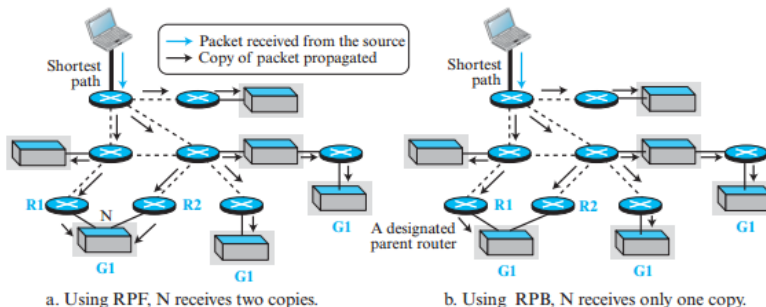
- RPF prevents duplicates at routers.
- RPB prevents duplicate copies from reaching the same network.
- One attached router is designated as the parent for a network for a particular source.
- The shortest-path router can be selected as parent; ties can use the smaller IP address.

Result

A shortest-path broadcast tree is formed with the source as root and networks as leaves.

Figure 4.96

Figure 4.96 RPF versus RPB



Reverse Path Multicasting (RPM)

- RPB broadcasts to all networks, including networks without group members.
- RPM prunes branches that do not lead to active members.
- IGMP provides membership information at the leaf level.
- Membership information is propagated upward through the reverse shortest-path tree.

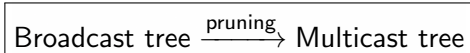
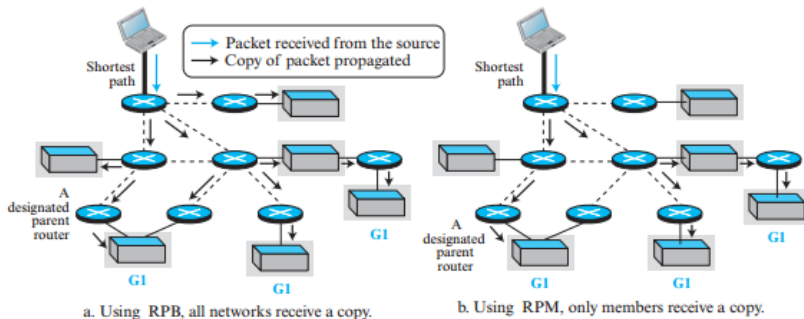


Figure 4.97

Figure 4.97 *RPB versus RPM*

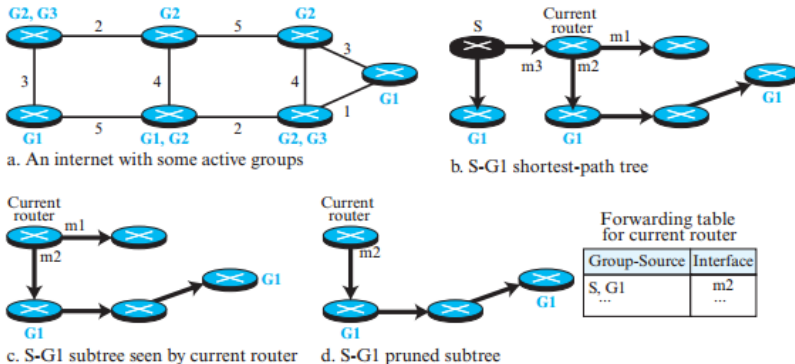


MOSPF – Multicast Open Shortest Path First

- Extension of OSPF using the source-based tree approach.
 - Router has an LSDB plus multicast group membership information.
 - Dijkstra is run with the multicast packet's source as the root.
- 1 Create source-rooted shortest-path tree.
 - 2 Find the router's subtree.
 - 3 Treat it as a broadcast subtree.
 - 4 Use IGMP/member information to prune branches.
 - 5 Forward through remaining multicast-tree interfaces.

Figure 4.98

Figure 4.98 Example of tree formation in MOSPF



Protocol Independent Multicast (PIM)

- PIM needs a unicast routing protocol for its forwarding table.
- The underlying protocol can be distance-vector or link-state.
- PIM has two modes based on group density.

PIM-DM

Dense mode; many routers have active group members.

PIM-SM

Sparse mode; relatively few routers have active group members.

PIM Dense Mode (PIM-DM)

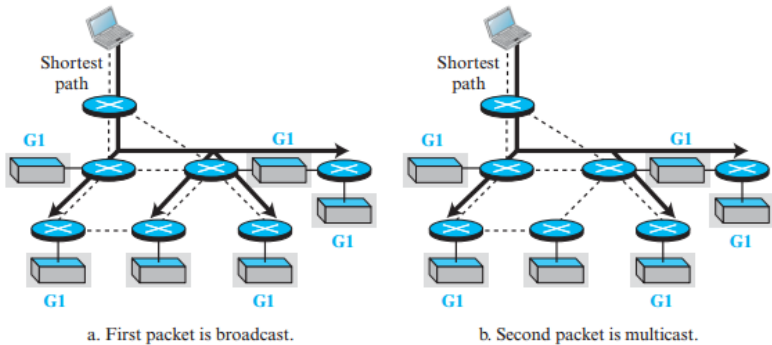
- ① Apply RPF to an incoming packet from source S for group G .
- ② Wrong incoming interface: discard and send prune.
- ③ Correct incoming interface: forward through eligible interfaces.
- ④ Routers without members send prune messages.

Characteristic

The first packet may be broadcast; pruning changes subsequent delivery into multicast.

Figure 4.99

Figure 4.99 *Idea behind PIM-DM*



PIM Sparse Mode (PIM-SM)

- Uses a group-shared tree.
- The core router is the **Rendezvous Point (RP)**.
- Source encapsulates a multicast packet in a unicast packet and tunnels it to the RP.
- RP decapsulates the packet and forwards it to the group.
- Each active group requires an RP, although a router can serve multiple groups.

PIM-SM – Join and Prune

Join

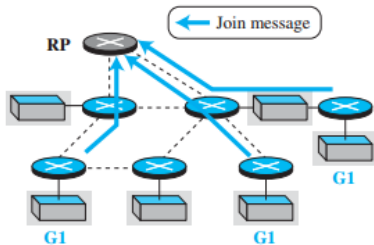
IGMP reports a new member. The designated router sends a join toward the RP. Routers along the path add the required multicast outgoing interface.

Prune

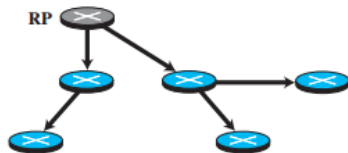
When no active member remains, the designated router sends a prune. Join count is reduced; when it reaches zero, the interface is removed from the multicast tree.

Figure 4.100

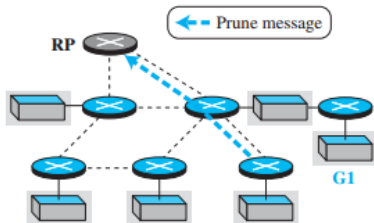
Figure 4.100 *Idea behind PIM-SM*



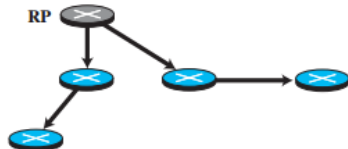
a. Three networks join group G1



b. Multicast tree after joins



a. Three networks join group G1



b. Multicast tree after joins

PIM-DM vs. PIM-SM

Feature	PIM-DM	PIM-SM
Mode	Dense	Sparse
Expected members	Many	Few
Tree approach	Source-based	Group-shared
Core/RP	Not required as group core	Rendezvous Point
Initial behavior	Flood/broadcast then prune	Join branches toward RP
Main control	RPF + prune	Join + prune

Interdomain Multicast Routing

- DVMRP, MOSPF and PIM are intended for multicast routing within an AS.
- When group members are spread across different ASs, interdomain multicast routing is required.

Protocols

- Multicast Border Gateway Protocol (MBGP)
- Multicast Source Discovery Protocol (MSDP)
- Border Gateway Multicast Protocol (BGMP)

MBGP – Multicast Border Gateway Protocol

- Extension of BGP for interdomain multicast routing.
- Provides separate paths for unicast and multicast.
- Exchanges multicast information between border routers.
- Uses a group-shared multicast routing approach.
- One router in each AS is selected as the rendezvous point.

MSDP and BGMP

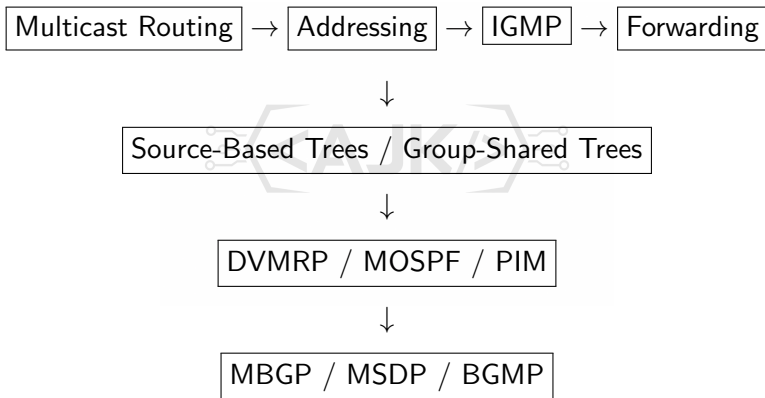
MSDP

A source representative router in each AS informs RPs about multicast sources available in that AS.

BGMP

Proposed as a possible replacement for MBGP. Builds a shared-group tree with one root AS for a group; sources must know how to reach the root.

Multicast Routing – Complete Concept Map



Multicast Routing – Key Takeaways

- Multicast is one-to-many communication using a group address.
- IPv4 multicast addresses occupy 224.0.0.0/4.
- IGMP collects group membership information.
- Multicast forwarding depends on source and destination.
- Source-based trees use one tree for each source-group combination.
- Group-shared trees use one core/RP tree per group.
- DVMRP uses RPF, RPB and RPM.
- MOSPF extends OSPF using source-rooted shortest-path trees.
- PIM operates in dense and sparse modes.
- MBGP, MSDP and BGMP address interdomain multicast routing.